

1. In one week in 2017, a technician earned a total of \$716 by working at her regular job and a second part-time job. The equation $20h + 14c = 716$ represents this situation where h is the number of hours worked at her regular job and c is the number of hours worked at her second job. Which of the following is the best interpretation of 20 in this context?
 - A. The amount, in dollars, the technician earned for each hour she worked at her regular job
 - B. The amount, in dollars, the technician earned for each hour she worked at her second job
 - C. The number of hours the technician worked in one week at her regular job
 - D. The number of hours the technician worked in one week at her second job

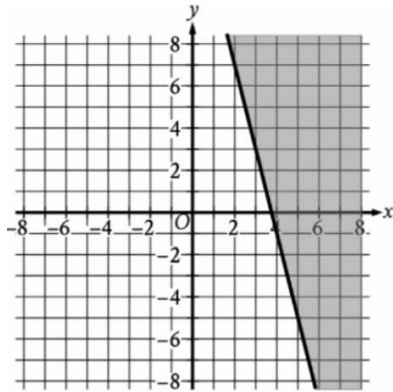
2. A marine biologist uses a linear model to estimate the weight of a blue whale after it is born. The model estimates that a certain blue whale weighs 5,710 pounds at birth and gains 10.0 pounds per hour, for 120 hours, after it is born. Based on this model, what is the estimated weight, in pounds, of this blue whale 7 hours after it is born?

3. $q - 34r = s$

The given equation relates the positive numbers q , r , and s . Which equation correctly expresses q in terms of r and s ?

- A. $q = s - 34r$
- B. $q = s + 34r$
- C. $q = 34rs$
- D. $q = -\frac{s}{34r}$

4. The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?



- A. $(-5, 0)$
B. $(0, 5)$
C. $(0, -5)$
D. $(5, 0)$
5. A rectangle has a length that is 49 times its width.
The function $y = (49w)(w)$ represents this situation where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $49w$ in this context?
- A. The length of the rectangle, in feet
B. The area of the rectangle, in square feet
C. The difference between the length and the width of the rectangle, in feet
D. The width of the rectangle, in feet
6. If $5(x + 4) = 4(x + 4) + 39$, what is the value of $x + 4$?
- A. -4
B. 35
C. 39
D. 43

7. The solution to the system of equations is (x, y) . What is the value of $2x$?

$$x + 10 = -9y + 5$$

$$x - 10 = 9y + 7$$

A. $-\frac{11}{9}$

B. 6

C. 12

D. 24

8. In 8 days, a polar bear ate 34.4 pounds of fat. Which equation describes the amount of fat, y , in pounds, the polar bear ate in these 8 days?

A. $y = 8$

B. $y = \frac{34.4}{8}$

C. $y = 34.4$

D. $y = 34.4 + 8$

9. To study fluctuations in leaf water potential, samples of wood were taken from 25 trees and cut in the shape of a cube. The length of the edge of one of these cubes is 3.000 centimeters. This cube has a density of 0.220 grams per cubic centimeter. What is the mass of this cube, in grams?

10. What are all solutions to the given equation below?

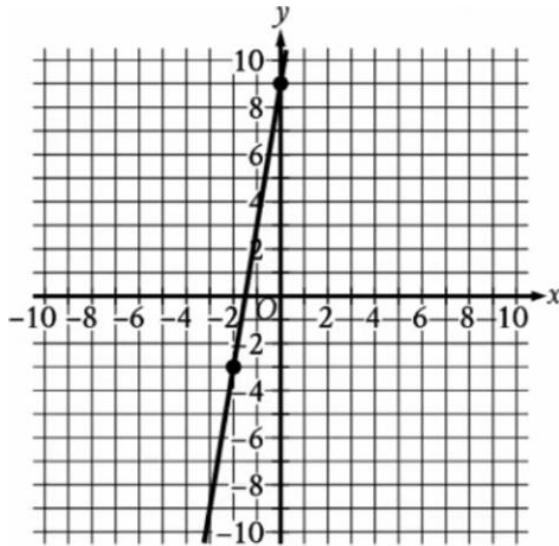
$$-4|5x + 6| + 4 = -20$$

- A. 0
- B. 0 and $-\frac{2}{5}$
- C. 0 and $-\frac{12}{5}$
- D. There is no solution.

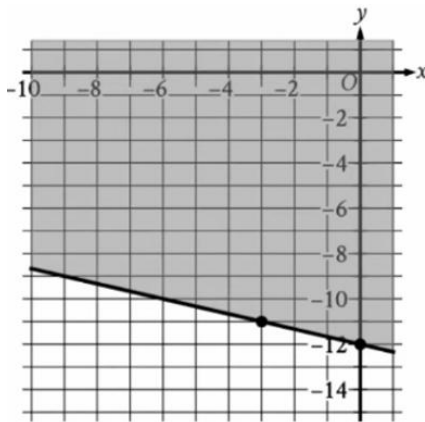
11. For the linear function p , $p(c) = -2$, where c is a constant, $p(4) = 38$, and the slope of the graph of $y = p(x)$ in the xy -plane is 8. For the linear function t , $t(c) = -3$ and $t(5) = 51$. What is the slope of the graph of $y = t(x)$ in the xy -plane?

- A. -1
- B. 6
- C. 8
- D. 9

12. The graph shows a linear relationship between x and y . Which equation represents this relationship, where R is a positive constant?



- A. $Rx + 12y = 18$
 B. $Rx - 12y = -18$
 C. $12x + Ry = 18$
 D. $12x - Ry = -18$
13. The shaded region shown represents the solutions to $rx + ty \geq -36$, where r and t are constants. What is the value of $r + t$?

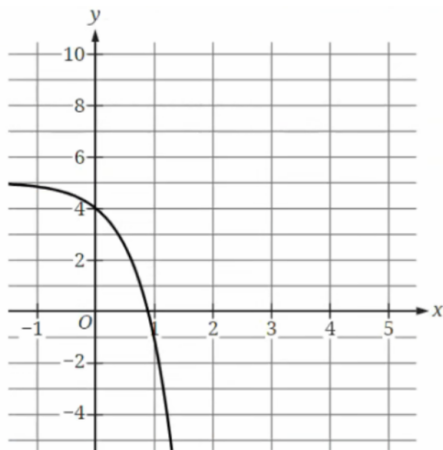


- A. 4
 B. 2
 C. -2
 D. -3

14. The edge length, in inches, of cube Y is $\frac{3}{86}$ the edge length, in inches, of cube X . The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X . What is the value of n ?

- A. $\frac{9}{7396}$
- B. $\frac{27}{3608}$
- C. $\frac{3}{86}$
- D. $\frac{9}{43}$

15. The graph of $y = f(x) + 2$ is shown. Which equation defines function f ?



- A. $f(x) = -6^x + 1$
- B. $f(x) = -6^x + 3$
- C. $f(x) = -6^x + 4$
- D. $f(x) = -6^x + 5$

16. Which expression is equivalent to $15(x + 10)$?

- A. $15x + 10$
- B. $15x + 150$
- C. $15x + 25$
- D. $15x + 5$

17. A square map has a side length of 35 inches, and 1 inch on the map represents an actual distance of 13 miles. A smaller version of the same map is printed as a square with the side length 70% shorter than the side length of the original map. On the smaller map, which of the following is the closest to the actual distance, in miles, represented by 1 inch?

- A. 3.9
- B. 7.65
- C. 24.5
- D. 43.33

18. In the given system of equations, c is a constant. The system has two distinct real solutions. Which of the following could be the value of c ?

$$y = x - c$$
$$y = -6(x - 12)^2$$

- A. 7
- B. 11
- C. $\frac{287}{24}$
- D. 17

19. In triangle ABC, the measure of angle A is 58° and $AC=30$. In triangle PQR, the measure of angle P is 58° and $PR=120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR?

- A. $AB=35$ and $PQ=35$
- B. $AB=35$ and $QR=140$
- C. The measures of angle B and angle R are 34° and 88° , respectively.
- D. The measures of angle B and angle Q are 58° and 34° , respectively.

20. The perimeter of an equilateral triangle is 876 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?

21. The function f is defined by $f(x) = \frac{|x|}{a} - 16$, where $a < 0$. What is the product of $f(17a)$ and $f(8a)$?

22. The function f is defined by $f(x) = a \cdot (b)^{\frac{x}{n}}$, where a , b , and n are constants, and b and n are integers. If $f(5) = 7$ and $f(8) = 189$, what is the value of $f(10)$?